

Assignment 3
CAP 6419 - 3D Computer Vision
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Q1. (20 pts) Using the transformation rules for points and lines show that two lines, $\mathbf{l}_1, \mathbf{l}_2$ and two points $\mathbf{m}_1, \mathbf{m}_2$, not lying on the lines, have the invariant

$$I = \frac{(\mathbf{l}_1^T \mathbf{m}_1)(\mathbf{l}_2^T \mathbf{m}_2)}{(\mathbf{l}_1^T \mathbf{m}_2)(\mathbf{l}_2^T \mathbf{m}_1)}$$

Q2. (20 pts) Using the transformation rules for points and conics show that a conic $\mathbf{C}$ and two points $\mathbf{m}_1, \mathbf{m}_2$, not lying on the conic, have the invariant

$$I = \frac{(\mathbf{m}_1^T \mathbf{C} \mathbf{m}_2)^2}{(\mathbf{m}_1^T \mathbf{C} \mathbf{m}_1)(\mathbf{m}_2^T \mathbf{C} \mathbf{m}_2)}$$

Q3. (20 pts) Show that an affine camera (i.e. a camera with a 3×4 projection matrix $\mathbf{P}$, whose last row is $[0 \ 0 \ 0 \ 1]$) maps parallel lines in the 3D world to parallel lines in the image. Show that this is not necessarily the case if the camera is not affine.

Q4. (20 pts)

- a. Show that the first three columns of a general projective camera matrix $\mathbf{P}$ correspond to the vanishing points along the three world coordinate axes up to unknown scales, i.e.
 $\mathbf{P} = [\mathbf{p}_1 \ \mathbf{p}_2 \ \mathbf{p}_3 \ \mathbf{p}_4]$, where $\mathbf{p}_1 = \alpha \mathbf{v}_x$, $\mathbf{p}_2 = \beta \mathbf{v}_y$, and $\mathbf{p}_3 = \gamma \mathbf{v}_z$
 (Hint: $\mathbf{v}_x, \mathbf{v}_y$, and $\mathbf{v}_z$, are the images of infinite points $[1 \ 0 \ 0 \ 0]^T$, $[0 \ 1 \ 0 \ 0]^T$ and $[0 \ 0 \ 1 \ 0]^T$, respectively.)
- b. Show that the fourth column of a general projective camera matrix $\mathbf{P}$ correspond to the projection of the origin of the world coordinate system into the image plane.

Q5. (20 pts) Let $\mathbf{v}_x$ and $\mathbf{v}_y$ be the vanishing points along the x and y axes, respectively. Prove that $\mathbf{v}_x^T \boldsymbol{\omega} \mathbf{v}_y = 0$, where $\boldsymbol{\omega} = \mathbf{K}^{-T} \mathbf{K}^{-1}$.